Touch, agency and the interplay between dance, disability and robotics

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Touch, agency and the interplay between dance, disability and robotics

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ABSTRACT

The themes of touch, contact and consent in relation to disability generate a lot of discussion and are brought further into view in the context of performance, and specifically dance performance. Disabled dance artists develop expertise that is particular to their own body, and if they dance with an assistive technology, such as a prosthesis, crutch or wheelchair, the dancer has additional expertise in how their corporeal body interfaces with non-human agents. If the non-human agent is a robot, questions around touch and consent may take on a different force. This article discusses a recent project that brought together an interdisciplinary team to ask questions about how expert disabled dancers encounter robots, and how bodily contact with robots can be generative and reimagine contact as creative, expressive and trustworthy, rather than potentially harmful. The discussion also asks whether AI offers an opportunity to develop more inclusive research methodologies, or whether it calls into sharp focus the inequalities if the lived experience of difference and diversity is overlooked.

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Introduction

The themes of touch, contact and consent in relation to disability generate a lot of discussion, often centring on issues around agency and autonomy (e.g. Dymoke 2019; Hammer 2024; Horrigan 2024). These issues are brought further into view in the context of performance, and specifically dance performance, where the body is 'on show' and physical contact is a regular aspect of the dancer's vocabulary. Disabled dance artists develop expertise that is particular to their own body, and if they dance with an assistive technology, such as a prosthesis, crutch or wheelchair, the dancer has additional expertise in how their corporeal body interfaces with non-human agents. However, the 'assistance' of technology also evokes the medical model of disability (Brisenden 1986), whereby the

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disabled person is regarded as deviating from what is considered a normally functioning body, and medical processes focus on providing medical interventions towards achieving 'normality'. Whilst disabled people may experience positive medical support throughout their lives, their lived experience is often one of being disabled by the wider environment built on normative structures, and many seek to argue against a deficit model, whereby disability may be viewed positively, not as loss, and difference is seen as 'normal', and the expertise developed in living differently in the world is respected. Hence there is a need to call out normative methods and methodologies that reinscribe the disabled body as deficient, transgressive, or unable. What emerges may be knowledge about the material relationship between a disabled dancer's own body and a mechanical device that becomes integrated in a way that poses questions about the experience of what is and isn't the body: the blurring between human and non-human. The 'queer animacy' (Belser 2016; Chen 2012) experienced by the dancers as the porous borders between their physical body and the prostheses can then be a creative source for dance making and performance.

If the non-human agent is a robot, which may *appear* to behave as if it has independent movement potential, questions around touch and consent may take on a different force. It is in this context that there is a growing interest in how dance can shape robotics, in both practice and research (Benford et al. 2024). Whilst appearing to have some autonomy (and which fuels anxiety about the rise of AI in relation to lack of human control), the apparent independent motion of the robot is most often a complex integration of AI, computing programming, robot wrangling, environmental considerations, and compliance with the manufacturers' regulations. These regulatory frameworks are vital for many reasons but may feel constraining when participating in an exploratory context and perhaps more particularly within a creative setting in which disabled professional dancers are not only acutely aware of risk to their safety in everyday living (for example, the risk of tripping or falling) but are often naturally 'risk takers', so it needs care. Health and Safety is thus an important consideration, and whilst research in this field often focuses on avoiding contact with robots, if contact is desirable, then moving safely with robots should be 'learned through iterative bodily experience' (Benford et al. 2025b), so contact may be reframed and negotiated as an important facet of human interaction (Benford et al. 2025a).

This article draws on this context to discuss a recent project that brought together dance practitioners with researchers across dance, computing science, robotics and disability studies. At the core of the project was a commitment to exploring the complex question of embodied trust, as reflected in the title of the project, *Embodied trust in TAS: robots, dance, different bodies*.[1] The questions included: What happens when expert dancers, and in this case disabled dancers who use prosthetics, wheelchairs or other assistive technologies, and thus have a deep embodied experience of the human/machine interaction, encounter robots? How does this expertise raise questions about bodily contact with robots – how can that contact be generative and reimagine that bodily contact as creative, expressive and trustworthy, rather than potentially harmful? What does 'embodied trust' mean? The project also prompted a further questioning of how the increasing presence of AI in many areas of our lives offers an opportunity to develop more inclusive research methodologies, or whether it calls into sharp focus the inequalities if the lived experience of difference and diversity is overlooked.

We align with researchers who have made the case for why robotics benefits from the knowledge and experience of dancers, as LaViers et al. argue:

> professional choreographers, dance practitioners, and movement analysts are critical to research in robotics. They have design methods for movement that align with human audience perception; they can help identify simplified features of movement that will effectively accomplish human-robot interaction goals; and they have detailed knowledge of the capacity of human movement. (2018, 1)

LaViers et al. point to the ways in which a dance perspective and sensibility contributes valuable knowledge of how movement is generated, how movement can be categorised for analytic purposes, and how movement transmits meaning. The perceptual properties of dance are important in acknowledging that dance is an aesthetic experience, capable of stimulating corporeal sensations in the viewer (Cross et al. 2011, 102). By incorporating the aesthetic and its relationship with a conscious awareness of bodily processes, Human Robot Interaction (HRI) may benefit from somatic-informed collaborations. This paper will describe how the project developed, primarily focusing on how the working practices evolved and what this might offer to collaborations between dance and robots, and to the broader field of HRI. Some of the early findings are already contributing to new enquiries, so whilst this first phase of the research is completed, it has led to additional questions and enquiries that will be mentioned lightly in the closing sections.

Materials and methods: the complexities of touch in HRI

There is a growing body of literature that is exploring the ways in which robots can be designed to be more human-like, and dance is often drawn upon in studies that attempt to build more animacy into the design. Some studies seek to emulate the dancer in the action of the robot. Others use dance as a site for examining how robots are influencing aesthetic preferences (Darda et al. 2024), whilst others may explore how robots can be designed to appear to be making their own decisions about how robots dance to music (Savery, Rogel, and Weinberg 2024). A number of studies are interested in embodied meaning-making in human–robot encounters, using arts-based methods to explore 'intra-bodily resonances' (Gemeinboeck 2021). Others are interested in how models of human motion can achieve perceptual imitation in robots (Kaushik and LaViers 2019). By contrast, some studies are interested in designing robots to dance more like humans (Rogel et al. 2022), or how AI and robots can imagine new kinds of dance elements and open new methods for dance creation (Yang 2022). A different strand of research, but which also has connections with our own questions, is positioned within social robotics (Law, Malle, and Scheutz 2021). Relatedly, research into digital intimacies raises questions about the ethics of consent (Ley and Rambukkana 2021), and Harvey and Drazek, in considering dance, intimacy and inclusion, 'suggest that moments of performed and experienced intimacy in dance/movement hold the potential to facilitate change' (2024). Questions of consent in HRI are often located within studies that address HRI in health contexts. As Lintvedt points out:

> The terms 'informed consent' and 'consent' are used interchangeably with an emphasis on informational privacy, whereas the embodiment of robots also intervenes with 'physical' privacy. This lack of nuance leads to misconceptions about the role consent can play in HRI. (2023, 1)

These studies are all advancing understanding about HRI, but what makes our study potentially novel is its exploration of how the direct physical interaction between expert dancers and robots can reveal more about the subtleties of what it means to touch and be touched. In consideration of the ethics that underpin human/robot connections, touch might also reveal new insights into what consent means when both the dancers and robots are interacting within a strange and unfamiliar environment and context (i.e. robots being made to behave in a way that they were not designed to do: dancers to perform in a robot environment subject to the controls, conditions, and regulations of a specialist space that is not the theatre studio). But consent is complex, touching on themes such as power dynamics, safety, risk and vulnerability. More specifically, the research team drew from the expertise of dancers who identify as disabled, acknowledging that whilst every dancer's body is unique, the dancers in our project have a specific expertise developed through many years of navigating the barriers that frequently inhibit their participation and shape the different demands on their lives. These demands may include how to reconcile the relationship between their corporeal body and non-human elements, how to claim expertise within somatics as systems of knowledge, and how to balance the demands of a career as a professional artist with the time needed to ensure their access rights are supported and what is often continual medical interventions. We thus employ an arts-based methodology rather than recruiting empirical psychological or cognitive methods. We use qualitative and somatic approaches with the aim to contribute new embodied and phenomenological insights.

When touch is implicated in HRI it may be aligned with questions around intimacy (Zheng et al. 2020) and studies of this kind often design the robot to have more physical gendered properties (Shiomi et al. 2018), and the extent to which interpersonal touch can be replicated in robots (Hoffmann and Krämer 2021). There are many studies, such as Hoffmann and Kramer, that focus on the social meaning of touch and how tactility is rooted in human need. Often these studies are focused on the difference between skin-to-skin contact in human–human touch, and skin-to-robot touch in HRI. In our study, the contact was almost entirely skin–to-robot touch, but we were interested in exploring how the confidence (or otherwise) in that contact was informed by the dancer's extra-daily (Barba 2021) experience of machine-to-skin contact (through the attachment of a prosthetic limb, for example). Moreover, touch may take on wider meanings not limited to making sense of part of the body touching another body part or something external. As phenomenologist Merleau-Ponty offers (1968), touch is both active (sentient) and passive (sensible), providing the example of a double sensation of touch when one hand touches the other hand. The touching hand considers the other hand as an object. The receiving hand feels the touch as a subjective experience. Merleau-Ponty's thesis on touch, written in 1968 before HRI and yet still influential in making sense of human experience, is situated within a normative view of what it means to touch and be touched, flesh to flesh. When the touch is flesh to robot, and the giver of that touch already has a sensation of simultaneously giving and receiving touch through being joined with a non-flesh 'object', the ambiguity of sentient and passive touch is made more complicated. Nonetheless, the dancers bring with them confidence in the value of touch, how touch is fundamental to their dance expertise, and how to respond to the sensations received through the touch of a non-human object (robot)

even if the relational aspects of touch are not attempting to replicate those experienced in a dancer-to-dancer context.

The dancers in the team included associate artists with Candoco Dance Company: Welly O'Brien, Kat Hawkins and Kimberley Harvey.[2] Over a series of workshops at the University of Nottingham's Cobot Maker Space, we embarked on a series of experiments moving with different robots in various configurations. We were committed to the principle of co-research and a co-design process whereby everyone is an equal team member (so the dancers were full team members, not research participants). Flattening hierarchies of knowledge that emerge from, or support power structures is not straightforward and needs time for working together in quite intense ways. At root is the importance of respecting all the expertise in the group and where different knowledges reside, as well as the systems or rules that play a necessary role in working practices. These practices can be related to safety protocols, which may present differently in the robotics and dance fields, or be familiar in one disciplinary field but new and sometimes even confusing in another. Other practices may be related to 'softer' processes that can sometimes be less visible or evident, such as how members of the group have sufficient space for physical preparation, where breaks are needed to take account of energy levels, and where 'time out' is needed to be away from the group. A community of practice (Wenger 1998) emerged through our sustained period of working together, but it needed regular 'check ins' to avoid assumptions about each other's needs, ways of working, and contributions. Core to our work was the importance of respecting each other's expertise, where knowledge resides, and the systems and rules that govern working practices. For example, the academic working culture can be quite different from the professional artist's working environment, so finding ways to bridge needs and expectations is important. We needed to respect the specialist space, how robots need to be used, how the space needs to be organised, and how time needs to be organised to support individual needs and provide space for a multivocal engagement. For example, we introduced the concept of what feminist crip theorists refer to as 'crip time' (Kafer 2013; Marsh 2023; Whatley 2023) to ensure that how time was structured respected the needs of everyone involved. It may be easy to assume that everyone knows each other's ways of working and the norms that permit particular activities so establishing an interdisciplinary culture, with no one practice, discipline or discourse feeling in service to another, needs work (Garrett et al. 2025a).

An important device for establishing the group involved beginning all our workshops by moving together to establish a common, shared ground from which the work developed (resisting codified or isolating spoken language). Led by team member, disabled dance artist and researcher Kate Marsh, the sessions developed from the principle of 'holding the space'. The explorations were disability-led, noting how everyone's 'normal' is different and fluctuating, enabling us to confront normative assumptions, language and challenging preconceptions. Our first workshop included a brief encounter with motion capture as we initially thought to include motion capture as a method for analysis, building on some earlier research (Whatley et al. 2023; Zhou et al. 2023), but the normative design of the motion capture suit (i.e. designed with the assumption of four limbs), the set-up of the working space, and lack of attention on tactility and the sensitivity of bodily contact was limiting, and we abandoned it as a method for this project.

Our workshops enabled us to 'play' with several different robots, and each sparked conversations about the extent to which they supported a process of partnering and offered insights into how we may ascribe human properties, including anatomical correspondences, functional aspects, emotions and even 'feelings' to the relationship with the robot, particularly those with human names. 'Play' is a familiar, often comfortable approach for dancers who are very experienced in improvisation but may not be comfortable for all in a context when the play with a robot is transgressing its purpose or limitations. If a robot is intended to provide a functional purpose and not be collaborative or touched, there will be the boundaries that should not be breached, but if breached, may produce a rupture that contributes new insights to HRI. Some robots felt more yielding or responsive to physical contact, or the potential for movement, or actual movement structures (such as rhythmic patterns, speed/trajectory through space, sounding) or felt closer to a dancer's own movement vocabulary. For example, O'Brien talked of finding an affinity with the telepresence robot due to its inherent instability and 'wobbliness' as it moved. Hawkins, by contrast, was drawn to 'Spot' the robot dog for its staccato and speedy gait. However, in these cases, there is a connection between the robot controller and dancer enacted through the robot: the robot is effectively a proxy being controlled by another actor, whether remotely driven or in close vicinity, evoking a puppeteer relationship.

Having tested the interplay with different robots, we decided to focus on one particular robot as it felt less predictive and more productive as a 'dancing partner': the Franka arm robot, with the caveat that the Franka arm is not designed for extensive human contact, so we needed to take care to observe the manufacturer's guidelines on where it is safe to touch (touch is only permitted at the switches at the wrist of the end effector when the robot is being operated).

The Franka arm robots may have human-like properties but when dancers moved with them, it was tempting to invest more human qualities, particularly as they were already (prior to our research) named 'Bob' and 'Vincent'. Hawkins' and O'Brien's dance involved a rich movement vocabulary of gesturing, folding, releasing, pausing, stretching, opening and so on, which transmitted to expand what resonated in the robot's more uni-directional and uni-dynamical movement. Having settled on the Franka arm we needed to test our own thinking about the extent we invested human properties in the robot. We tried out several experiments. We named parts of the robot as we perceived them to suggest human body parts (labelled sometimes anatomically or more imaginatively), or we ascribed feelings or expressions to different robot parts, expanding the naming to more poetic descriptions and reflections, partly prompted by how we had seen them move and 'being moved'. There was some consensus amongst the team but also many different interpretations. Although the naming of the robots appears to gender them as male, reflecting what at least one study has found that 'robots with a higher degree of anthropomorphism were perceived as significantly more male than neutral or female' (Roesler, Heuring, and Onnasch 2023), there was little in our exercise that confirmed a gendering of the robots; although further research may consider how gender plays a role in robot design and interaction. Nonetheless, the working process seemed to build both a closer relationship with the robots and between the team as we sought common ground in our human interaction and engagement.

Concepts/conceptual framework

As the workshops progressed, we developed a conceptual framework derived from four core themes that emerged through the practice and acted as a reference point for the explorations. The first theme focused on bodies and our premise that all bodies are plural, gendered, vulnerable and plastic. The next theme centred on how control is related to agency, consent, trust and responsibility. The third explored how translation is core to both motion and language. The final theme, contact and collision, emerged from being mindful of the inherent complexities of collision as a creative stimulus within HRI whilst recognising the considerable research into collision detection in robotics. These themes, evolved to conceptualise the experimental work during the project, formed a basis for developing working protocols, but it is important to note that the themes are interconnected, so in practice, they operated more as a meshwork (Ingold 2011, 63–94) where everything tangles with everything else and continues to shift and move through the practice, and where coherence is found through contrary forces of tension and friction (Ingold 2017, 10). To follow Ingold's thinking, the process came into being through inhabiting an emerging and transformational working process. The framework was thus helpful for articulating the principles that helped with guiding the activity and which supported reflective discussions, and are expanded in Figure 4.

The conceptual framework and the early experiments supported and informed the devising of several scenarios to support different guided dancer/robot improvisations as a basis for choreographic sketches. The scenarios were simple task-like improvisational scores or guides for the dancers to explore with the robots.

Setting the space

As we were working in the Cobot Maker Space, a space set up for robotics research rather than dance practice, we needed to consider how to construct a safe and creative working space. Health and Safety considerations were very important. The robots needed to be programmed so that they would be able to move in relation to the dancers but in a safe way, noting that the two dancers involved at this point of the research, Hawkins and O'Brien, required different support for their participation. O'Brien is a single limb amputee who prefers to dance without her prosthesis. Hawkins, as a double lower limb amputee, chose to move sometimes with their prostheses and sometimes without. Their decisions to perform with or without their prostheses are sometimes based on artistic choices and sometimes are driven by a more pragmatic consideration based on the comfort, or lack of comfort, of wearing the prosthesis at different times. The working space was thus set up for practical considerations, both in terms of safe practice and to support how Hawkins and O'Brien would be comfortable in the interaction with the robots (see Figure 1).

The Franka arms were securely mounted on tables, facing each other and mirroring one another in opposition. A rostrum was placed next to one robot so O'Brien could dance in an elevated position next to the Franka arm. As a signifier of disability, together with Hawkins's prostheses when in the visual field, the set-up carried meaning, challenging normative assumptions that bodies are upright, can stand next to a robot, and

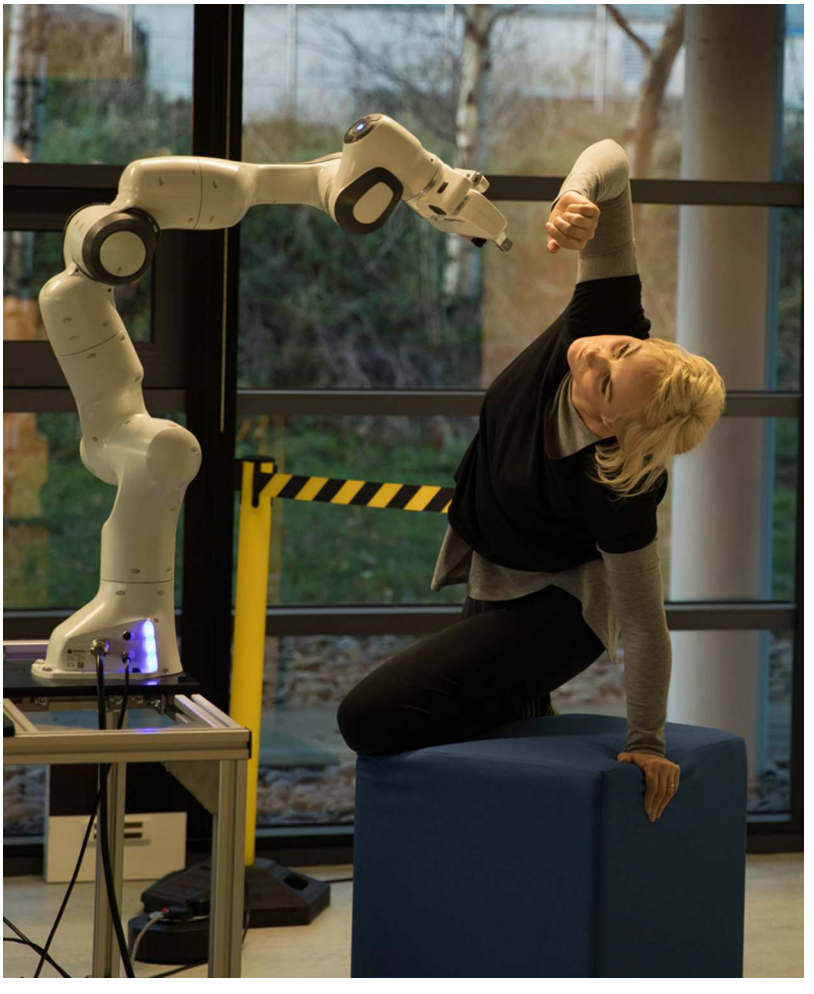


Figure 1. Welly O'Brien improvising with the Franka arm robot.

have four limbs that are able to move and push around the joints of a robot. The 'red button' was controlled by two 'spotters' standing behind the tables with the responsibility to stop one or the other robot if there was concern about the safety of the dancers (see Figure 2). Feng Zhou was sitting close by, so he was able to programme, record and have another level of intervention in the robots if safety was compromised. Simon Castle-Green was also closely observing from a safety point of view. The rostra for stability, together with the safety mechanisms, thus produced various signifiers of trust, amplified by Marsh directing the dancers from the front and providing music as a mood ground for the improvisions. The rest of the team was viewing as if an audience.

The robots ran in a teleoperation mode in which directly manipulating one would cause the second to mirror its movements in real-time, with one robot being designated a leader and the other the follower. Each Franka Panda robot was directly connected to its own dedicated PC, which was running the robot control software, which utilised the libfranka software,[3] a C++ library that provides low-level control of the Franka Robots. TCP



Figure 2. The set-up, with dancers Welly O'Brien and Kat Hawkins, spotters (Sarah Whatley and Rachael Garrett) and programmer (Feng Zhou).

sockets were created between the robots' dedicated PCs allowing the communication of movement instructions between them.[4] In leader mode, using libfranka, the robots' joint state was captured and transmitted via the TCP socket. In follower mode, the software receives the desired joint state and using libfranka, updates the robot to match the desired state. On the leader robot, we also saved joint states to a local data file which could be replayed at a later date using a playback mode. In this scenario, the playback software sends data over the TCP socket from the local file instead of from the physical robot, while the follower robot processes the data packets as normal (see Figure 3).[5,6]

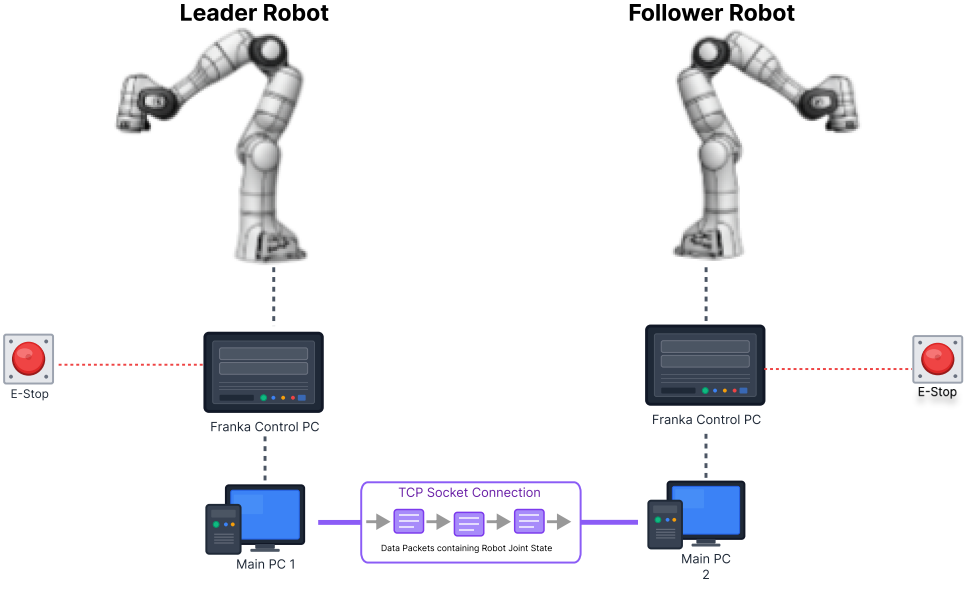


Figure 3. The schematic of the setup.

This setup, which created an unconventional theatrical setting, created a *mise-en-scène* that supported the choreographic structure, which in turn supported a wider ensemble of players/actors. Those implicated in the experiments became part of the *mise-en-scène*, so all who were in view took on a performative function. The human team members thus all moved between multiple roles aligned with the theatrical world, as performer, director, choreographer, prompt, stage manager, audience, intimacy coach, whilst also ensuring that the necessary technical, and health and safety procedures were active.

Performing the scenarios

We now outline some of the scenarios, or in some cases, more detailed scores that were devised to test out different connections between the dancers and Franka arms, and then describe one in more detail to discuss the observations and themes that emerged. Although all the explorations were improvised, the repetition of some scenarios led to the potential for choreographic sketches to form. Scenarios included one dancer as 'leader', moving the robot in response to several prompts (qualitative or descriptive words) from the rest of the team; the other dancer followed the leader, moving the robot with the aim to mirror the partnership of the lead. Another scenario involved O'Brien or Hawkins moving on their own with the robot, controlling its movement, which was then recorded as a set of coordinates in space. The robot then 'replayed' the duet as a solo without a dancer. The pre-programmed 'score' produced a different iteration of the original dance, demonstrating the potential for the robot to generate a sequence that had some quality and echo of the dancer, or could be re-embodied by the dancer. A variation of this scenario explored how the dancers could dance with their own pre-recorded robot or could improvise with their robot in response to the other dancing with their pre-programmed robot, or one dancer's recorded pathway was programmed on both robots and the dancers improvised in response to their 'own' robot' moving through the pre-programmed route. We played with different ways of recording or pre-programming the Franka arms to explore how they altered the felt connection between dancer and robot, robot and robot, or dancer to dancer. In some instances, the dancers were invited to describe their experiences as they were improvising a score and interacting with the robot.

Other scenarios involved focusing specifically on touch and one involved O'Brien moving in contact with the robot with her eyes closed. In this scenario, Hawkins was improvising with the other robot, which was programmed to replicate the movement of O'Brien's robot. The intention was not to stimulate visual impairment but to test out how trust, responsibility and agency were experienced in the context of dancing with a focus on tactility and light touch with the robot. O'Brien talked about how having her eyes closed meant relying on touch and memory of her relationship with the robot, her trust in how the robot moves, and trust in her partner. O'Brien described feeling 'completely safe' and 'most relaxed' because of a sense of not having 'responsibility' for the robot, and yet she was controlling the action: because she knows the robot, she knows where she is with it. O'Brien also described feeling able to sense the movement better with her eyes shut, that the relationship was more human with her eyes shut; the sensorial impulse was clearer. O'Brien referred to her 'whole system' responding to the touch, finding she was able to engage the rest of her body, and how the 'feeling' reverberated

through her whole body and she was less 'handy', implying she is less concerned with where to put hands through having gained expertise in where to touch. But by being bodily closer to the robot, O'Brien described having more control in the interaction, being able to 'feel it differently' without the distraction of looking elsewhere if her eyes had been open. O'Brien's experience seemed to chime with sociologist Richard Sennett's claim that 'the brain receives more trustworthy information from the touch of the hand than from images in the eye' (2008, 150).

Hawkins's improvisation in the same scenario seemed to convey the awareness of being the 'eyes' so a sense of care and sensitivity was present. Hawkins performed a bodily awareness of the risk inherent in the scenario. Hawkins also appeared to be testing the limits of what was acceptable risk, choosing, for example, to move under the Franka arm even if knowingly beyond its reach space. Hawkins talked about being willing to be in a less 'safe' place, but through building embodied knowledge about the kinaesthetic sphere of the robot, was able to draw on proprioceptive awareness to know where the limits lay. Both dancers acknowledged that their own individual spatial awareness was constructed through their relationship with a stick, a chair, or other assistive technology, which then played a role in how they related to the robot and mitigated risk by 'knowing' the robot, and their own dance, even if always in 'risk mode'. They also acknowledged that their pain threshold is generally high due to different medical interventions, although, as O'Brien observed, the media appearance of the Franka arm reminded her how disabled people are 'always looked at, always under a medical microscope'. Whilst there was a clear sense of confidence in the dancers' interactions with the robots, the sense of risk was transmitted to the spotters and watchers, heightening the tension felt by the spotters.

These sketches produced a very large corpus of video and image material, which have formed the basis for ongoing analysis and discussion. The collection is a loosely categorised archive of improvised danced scores. The intention is to consider how this content might be used for developing AI models that could generate new robot movements.

Considering consent

Our discussions centred on the experience of being 'in' the improvisation, in different roles. Several topics emerged relating to trust, autonomy, responsibility and how contact and interaction when improvising with the robots leads to more complex questions about consent, care, the ethics of touch, and whether notions of care and avoiding harm get transferred to the robot. As noted earlier, consent is a complex idea when involving robots and, importantly, is continuously negotiated by all parties. Whilst the dancers consented to explore dancing in partnership with robots and to drive the robots for their movement data to be collected for the purposes of programming the robots, consent resides within the interrelationships established through the research but it cannot be reciprocated by the robots. But this may not mean that the dancers are not moving with respect for the robot and its limitations, as if consent were a shared agreement and the robot had given permission to be touched. Consent is both obtained from those who own and operate the robot, even if via the robot, and it relies on a mutual understanding between the partners (human–human or human–robot) and where it is acceptable to touch.

Moreover, the explorations revealed how consent is continuously negotiated. Humans are in constantly changing states, depending on changing moods, a sense of wellness, or the social or working context (which may vary when in rehearsal or in public, or when working with different partners). These changing states have an equivalent in the robot; it may be happy to run one programme but not another. In our own observations, when the button was pressed to move the Franka arm it was much more responsive to and tolerant of gross movement than when it was running the mirroring programme. This then leads us to question if dancers can negotiate consent somatically, and just through 'known' or 'felt' social rules and experience, how might or should this work with robots. Is the robot a dancer, prop, extension to the choreographer, or the programmer? Who exactly is consent being negotiated with?

For the dancer, touch is fundamental to how they develop sensorial awareness with a dancing partner. Touch has different qualities, whether feathering, skimming, guiding, manipulating or colliding. But human bodies may be soft, fleshy, imperfect and responsive to heat, pressure, pain and so on. Robots are generally manufactured to be hard, often smooth, shiny and durable, yet can be vulnerable to heat, pressure and too much exertion on their joints.[7] Whilst human bodies are often able to express potential or actual discomfort, robots are not yet able to signal impending injury. But a shared sense of avoiding pain and injury, as if the dance were consensual, was negotiated through touch to ensure that how the dancers touched the robots was not going to harm themselves or damage the robot. The somatic experience of hurt, damage and even pain was transferred to the robot most clearly in the interaction with the red button, as the principal (and visible) safety device.

The red button, lock, glitch, or refusal to consent

As noted earlier, part of the safety procedure was to place two spotters close to the action with their hands on an emergency button, ready to press it to immediately stop the robot they were observing if there was a perceived danger to the dancer, to the robot or the environment. We were aware that using the Franka arms meant being aware of its limitations and how our interest in exploring different scenarios could lead to mistakes in presuming what the robot would 'do'. For example, whilst the robots were opposite each other and appeared to have a mirroring effect, albeit mirroring in opposite, it was not a mirrored environment; the raised rostrum on one side led to a different context in which each arm would replay a pathway if it was programmed on the other arm. So, when swapping which robot would 'perform' the pre-recording without swapping the dancer and the staging, it meant that playback of the prerecorded movement had less space than the original recording robot: the original robot had a larger 'safe' kinesphere in which it could move than the playback robot. This led, on one occasion, to a potentially dangerous collision between O'Brien and the robot, and the red button was pressed.

We noted the significance of this moment and its impact on the whole team, leading to discussion about responsibility, agency, acceptable risk, as well as noting how the person who pressed the red button experienced it as taking away control from the dancer. Those charged with 'hitting the button' were often anxious about intervening and imposing their anxiety to stop the robot's movement when the dancer's actions were perceived as intentional. What amounts to a shared responsibility is important, but challenging.

The robot may be stopped by an 'outsider' but may also stop itself if its joints are over-extended or over-contracted, as a self-protection mechanism. In line with the discussion on consent, there was a shared understanding between all parties that anybody involved in the practice can halt it at any time. As the dancers gained expertise in interacting with the Franka arms, they started to get a sense of the robot's limits and reported being able to 'feel' when it was getting close to locking, but it was almost always 'too late'. The lock is experienced as an abrupt freeze, or a glitch in the system, where the robot refuses to move any more. Each arm might respond differently due to its environment (based on having more or less free space around it) and might be perceived differently, as a sudden drop in energy, or an exaggerated resistance. The moment is experienced bodily by the dancer, ranging from curiosity to frustration, disappointment or even panic. But as the dancers gained more expertise in their interactions with Bob and Vincent, a process of mechanical sympathy developed, described by La Delfa as a process of sensing and acting that leads to a cumulative appreciation of human-with-machine (2023). The glitch may have heightened a mechanical sympathy for the dancers by relating to the glitch as the robot expressing what is a familiar condition of unsmooth bodily movement, broken time or 'crip time', whereby 'rather than bend disabled bodies and minds to meet the clock, crip time bends the clock to meet disabled bodies and minds' (Kafer 2013) and as Hawkins noted; 'glitch is part of the beauty of disability'. Increasing awareness of the robot's potential to embody similar unpredictable motion enabled them to find ways to 'act together' (La Delfa 2023). Each lock resulted in the need to return the robot to its 'safe' starting place, but with a lingering concern that avoiding the lock, and fixing the glitch, might imply that the disabled body also needs to be 'fixed'. However, the dancers' growing expertise in reacting to the glitch may expand our understanding of how robots are designed to relate to human bodies (Garrett et al. 2025b).

Results: future applications and a methodological schema

The work in this project is seeding new research that continues to explore how HRI is informed by the expertise of dance artists, and particularly dance artists who identify as disabled and have experience of prosthetics and/or assistive technologies. The work poses questions about trust, autonomy and how interaction and 'dancing with' robots reveal underlying assumptions and alternative possibilities concerning somatic relationships to robotic systems, and more about consent, care, and the ethics of touch. The interdisciplinary process has enabled us to begin building a shared discourse that crosses dance, HCI, soma design and robotics. Future work may return to motion capture as a mode of data capture and movement analysis to explore how human and non-human agents can be tracked and the subtleties of touch can be visualised in different ways.

A core theme throughout the project is how touch is experienced by, with and through the contact between dancer and robot. As experts in the affordances of 'touch' in performance, the dancers bring with them confidence in the value of touch, how touch is fundamental to their dance expertise, and how to respond to the sensations received through the touch of a non-human object (robot), even if the relational aspects of touch are not attempting to replicate that experienced in a dancer-to-dancer context. All the dancers talked about the relationship between touch and weight-giving, noting

how when wearing a prosthesis there is a particular experience of 'where' to give weight and how dancing with a robot heightens the sense of giving/taking weight when prosthetics are involved.

The conceptual framework that emerged through the project led to us identifying how the collective working practices constructed an interdisciplinary methodological schema, acknowledging how we bring different methodologies, epistemologies and ontologies into the research. Although there were times when different stages overlapped or were even skipped, depending on the group's focus at the time, the schema developed mostly sequentially as follows:

Stage 1 – Arriving. This stage involved creating a safe space for moving together, including checking in on expectations, safe practice and relevant health and safety/risk issues.

Stage 2 – Encountering. Through the close study of the robots, we described our impressions and collected descriptions based on close observations and before touching or moving them.

Stage 3 – Contact. As a core stage in the process, this stage focused on haptics, exploring making contact and engaging with the robots, noting the temperature, texture, weight, mass, and feel on the body. We experimented with the different processes of listening and following; leading and controlling; and added new observations based on these physical interactions. This is where we also considered how somadesign principles might influence the haptic and sensorial (somatic) relationship between the body and the robot.

Stage 4 – Translation/transformation. This stage involved exploring how the robots and the dancers are altered, influenced or changed by the interaction. We explored what motion the robots may stimulate in the dancers, such as to quiver, stretch, slide, fold, and how the robots consequently respond and move differently. We explored the limits of motion, the 'breaking points' experienced through locking or glitching, and whether the dancers and/or the robots change form and function under different conditions.

Stage 5 – Roots and connections. In conjunction with the practical explorations, we considered the historical, cultural, mechanical and functional roots of the robots and how those roots may align with the cultural, social, aesthetic and political roots of the dancers' practice to explore where there were intersections and divergences.

Stage 6 – Generation. Whilst the project wasn't primarily focused on creating performance outcomes, this stage involved the creation of scenarios that would generate improvisation in response to 'scores', building movement material around the emerging themes, emphasising the connections across dance, body and robot/s. This stage included discussing how other interacting materials such as Soma Bits, and collaborating components such as sound, lighting, location, and digital processes of documentation, including motion capture, may influence the collaboration. Choreographic sketches may then be built around significant or recurring themes that emerged through the improvisation scenarios.

Stage 7 – Analysis. Different modes of analysis focused on the distillation of the data as an iterative process of meaning-making. Processes such as tagging video recordings and audio recordings of the dancer/robot interaction led to identifying correspondences

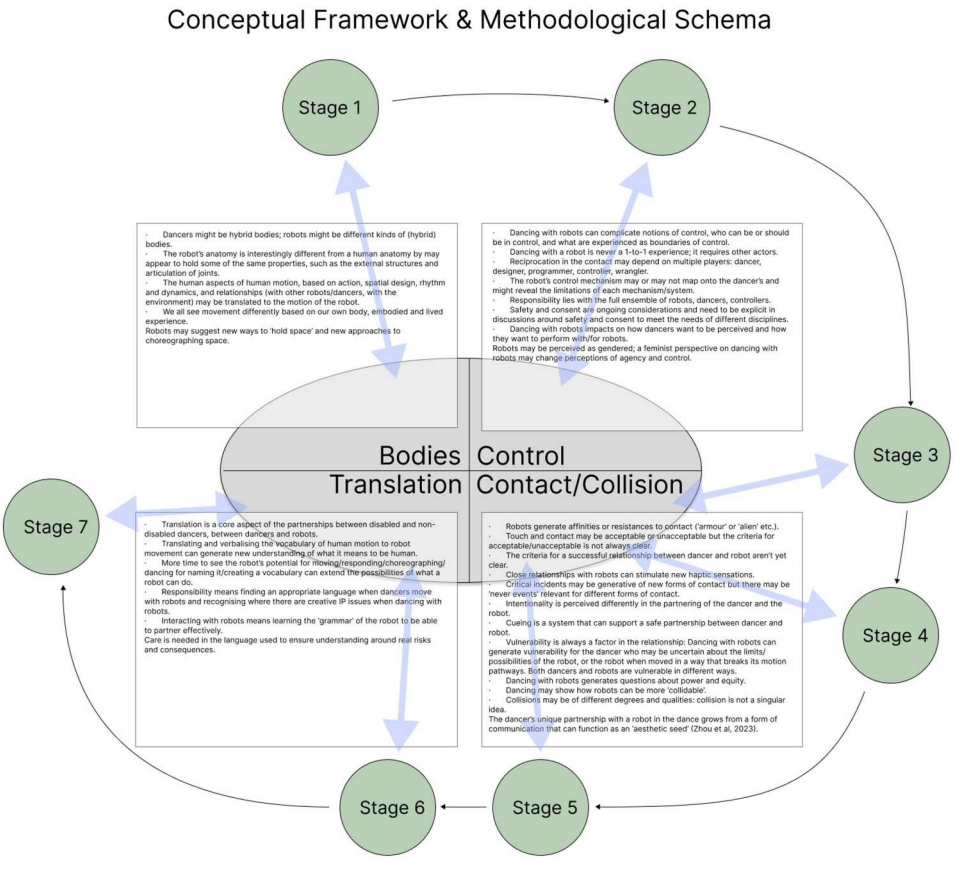


Figure 4. Conceptual framework integrated with a methodological schema.

between key terms, images and qualities that seemed to recur throughout the experiments. This final stage sought to reveal new insights and capture the interests that were rising through the collaborative process, including articulating the most compelling questions and interests to pursue, and what shared discourse was emerging.

The schema, which is illustrated with its interaction with the conceptual framework (see Figure 4), thus embodied the working methods and the logics of the composite narratives that were emerging through the intersection of dance, disability, robotics and prosthetics.

Summary

The project has led to findings that may be valuable for knowing more about the ways in which creative and performance methodologies can feed into robotics and technological design and methods, and may produce new kinds of performance scores and materials. The methodological schema described above could be reproduced, developed or adapted and together with the conceptual framework, may be applied to develop new directions and challenges for HRI research to improve the design of robots, including as dancing partners.

Findings include the following:

– Awareness of the influence of the environment in which robots are active.
– The benefits of recording and replaying movement interactions and the value of using the data to generate new movement explorations.
– New approaches to safety, risk and vulnerability.
– Softer and better-communicated boundaries for approaching freezing in robots.
– Methods from HRI and transdisciplinary collaborations can feed into creating new forms of performance practice.

In sum, we have identified ways in which the robot design process may benefit from dancers' expertise and lived experience, unveiling something about the robot's inner workings to show how 'cripping' established design processes may lead to new insights about risk, vulnerability, agency, and processes that support robot users. The learning could inform methods to help users of robots feel more comfortable in their interaction, enabling robots to be accessible to more people. Reciprocally, dancers noted how the work helped to 'tune' their proprioception; they expanded their movement vocabulary through duetting with robots and developed more expertise in mechanical sympathy through their somatic understanding of the interface between body and machine. From a technological perspective, the safety strategy performed by the robot under the control of the programmer or wrangler requires the dancer or user to align themselves with the safety system. But our experiments point to ways to improve the interactive potential of this system, whether based on sensors or other forms of robot 'feedback', including the lock. Knowing more about the robot's limitations could be beneficial for a range of applications that are being developed through AI. Our findings are informing new research enquiries that will continue to explore the intersection between dance, disability and robotics, to rethink normative methods of data capture, extraction and representation, and to consider what a 'body' means in techno-somatic environments.

Notes

1. The project ran from 2023 to 2024 as a TAS (Trustworthy Autonomous Systems) funded pump-priming project https://tas.ac.uk/research-projects-2023-24/embodied-trust-in-tas-robots-dance-different-bodies/.
2. In addition to the dance artists, the team included Sarah Whatley (lead), Steve Benford, Feng Zhou, Kate Marsh, Paul Tennent, Simon Castle-Green, Praminda Caleb-Solly, Kristina Höök, Rachael Garrett, Patrick Brundell, Juan Martinez Avila, Virginia Portillo.
3. https://github.com/frankarobotics/libfranka (verified 9th December 2025).
4. We gratefully acknowledge the support of Dr. Joseph Bolarinwa who developed this control software.
5. https://github.com/frankarobotics/libfranka (verified 9th December 2025).
6. We gratefully acknowledge the support of Dr. Joseph Bolarinwa who developed this control software.
7. We experimented with Soma Bits to find ways to soften the robot. Soma Bits were developed by KTH Royal Institute of Technology, Stockholm. They are a 'prototyping toolkit that facilitates Soma Design. Acting as an accessible "sociodigital material" Soma Bits allow designers to pair digital technologies, with their whole body and senses, as part of an iterative soma design process' https://www.kth.se/hct/mid/research/interaction-design/projects/soma-bits-1.921861.

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Kate Marsh is a disabled dance artist-researcher with over 20 years of experience in performing, teaching and making. Her interests are centred around perceptions of the body in the arts and notions of corporeal aesthetics. Specifically, she is interested in each of our lived experiences of our bodies, and how this does (or doesn't) inform our artistic practice. Her practice-research focusses on leadership in the context of dance and disability and draws strongly on the voices of artists to interrogate research questions.

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References:

Barba, E. 2021. "Dancing Questions." *Journal of Theatre Anthropology* 1:7–11.
Belser, J. W. 2016. "Vital Wheels: Disability, Relationality, and the Queer Animacy of Vibrant Things." *Hypatia* 31 (1): 5–21. https://doi.org/10.1111/hypa.12217.
Benford, S., R. Garrett, C. Li, P. Tennent, C. Núñez-Pacheco, A. Kucukyilmaz, V. Tsaknaki, et al. 2025a. "Tangles: Unpacking Extended Collision Experiences with Soma Trajectories." *ACM Transactions on Computer-Human Interaction* 32 (4): 1–34. https://doi.org/10.1145/3723875.
Benford, S., R. Garrett, E. Schneiders, P. Tennent, A. Chamberlain, J. Avila, P. Brundell, and S. Castle-Green. 2024. "How Artists Improvise and Provoke Robotics." In *International Conference on Social Robotics*, 66–77. Singapore: Springer Nature Singapore.
Benford, S., E. Schneiders, J. P. M. Avila, P. Caleb-Solly, P. R. Brundell, S. Castle-Green, F. Zhou, et al. 2025b. "Somatic Safety: An Embodied Approach Towards Safe Human-Robot Interaction." In *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, 429–438. IEEE.
Brisenden, S. 1986. "Independent Living and the Medical Model of Disability." *Disability, Handicap & Society* 1 (2): 173–178. https://doi.org/10.1080/02674648666780171.
Chen, M. Y. 2012. *Animacies: Biopolitics, Racial Mattering, and Queer Affect*. Durham and London: Duke University Press.
Cross, E. S., L. Kirsch, L. F. Ticini, and S. Schütz-Bosbach. 2011. "The Impact of Aesthetic Evaluation and Physical Ability on Dance Perception." *Frontiers in Human Neuroscience* 5:102. https://doi.org/10.3389/fnhum.2011.00102.
Darda, K M, A Maiwald, T Raghuram, and E S Cross. 2024. "Dancing Robots: Aesthetic Engagement is Shaped by Stimulus and Knowledge Cues to Human Animacy." *Frontiers in Human Neuroscience* 18: 1413066. https://doi.org/10.3389/fnhum.2024.1413066.
Dymoke, K. 2019. "Touching the Untouchables." *Psychotherapy and Politics International* 17 (3): e1506. https://doi.org/10.1002/ppi.1506.
Garrett, R., P. Brundell, S. Castle-Green, K. Hawkins, P. Tennent, F. Zhou, A. Lampinen, K. Höök, and S. Benford. 2025a. "Friction in Processual Ethics: Reconfiguring Ethical Relations in Interdisciplinary Research." In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 1–15.
Garrett, R., K. Hawkins, P. Brundell, S. Castle-Green, P. Tennent, F. Zhou, A. Lampinen, K. Höök, and S. Benford. 2025b. "In the Moment of Glitch: Engaging with Misalignments in Ethical Practice." In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, 1–18.
Gemeinboeck, P. 2021. "The Aesthetics of Encounter: A Relational-Performative Design Approach to Human-Robot Interaction." *Frontiers in Robotics and AI* 7:577900. https://doi.org/10.3389/frobt.2020.577900.
Hammer, G. 2024. "Navigating Frictions: Crip Utopian Performatives in Integrated Dance." *Etnofoor* 36 (2): 109–130.
Harvey, K., and R. Drazek. 2024. *We Suggest that Moments of Performed and Experienced Intimacy in Dance/Movement Hold the Potential to Facilitate Change*. Accessed January 28, 2025. https://jdspedi.wordpress.com/2024/09/06/dance-intimacy-and-inclusion/.
Hoffmann, L., and N. C. Krämer. 2021. "The Persuasive Power of Robot Touch. Behavioral and Evaluative Consequences of Non-functional Touch from a Robot." *PLoS One* 16 (5): e0249554. https://doi.org/10.1371/journal.pone.0249554.
Horrigan, K. 2024. "Risky Dancing and Consent Practices: Addressing Identity, Privilege and Power in Contact Improvisation." *Journal of Dance & Somatic Practices* 16 (1-2): 91–106. https://doi.org/10.1386/jdsp_00122_1.
Ingold, T. 2011. *Being Alive: Essays on Movement, Knowledge and Description*. Abingdon: Routledge.
Ingold, T. 2017. "On Human Correspondence." *Journal of the Royal Anthropological Institute* 23 (1): 9–27. https://doi.org/10.1111/1467-9655.12541.

Kafer, A. 2013. *Feminist, Queer, Crip*. Bloomington: Indiana University Press.
Kaushik, R., and A. Laviers. 2019. "Imitation of Human Motion by Low Degree-of-Freedom Simulated Robots and Human Preference for Mappings Driven by Spinal, Arm, and Leg Activity." *"International Journal of Social Robotics* 11: 765–782. https://doi.org/10.1007/s12369-019-00595-y.
La Delfa, J. 2023. "Cultivating Mechanical Sympathy: Making meaning with ambiguous machines." Doctoral dissertation, KTH Royal Institute of Technology.
LaViers, A., C. Cuan, C. Maguire, K. Bradley, K. Brooks Mata, A. Nilles, I. Vidrin, et al. 2018. "Choreographic and Somatic Approaches for the Development of Expressive Robotic Systems." *Arts* 7 (2): 11. MDPI. https://doi.org/10.3390/arts7020011.
Law, T., B. F. Malle, and M. Scheutz. 2021. "A Touching Connection: How Observing Robotic Touch Can Affect Human Trust in a Robot." *International Journal of Social Robotics* 13 (8): 2003–2019. https://doi.org/10.1007/s12369-020-00729-7.
Ley, M., and N. Rambukkana. 2021. "Touching at a Distance: Digital Intimacies, Haptic Platforms, and the Ethics of Consent." *Science and Engineering Ethics* 27:1–17. https://doi.org/10.1007/s11948-021-00338-1.
Lintvedt, M. N. 2023. "A Critical Analysis of Consent in Human–Robot Interaction." In *Cambridge Handbook on Law, Regulation, and Policy for Human-Robot Interaction* (forthcoming), University of Oslo Faculty of Law Research Paper (2023-01).
Marsh, K. 2023. "Crip Time and the Creative Process: Choreography and Performance as Sites for Exploring "Normative" and "Crip" Time and the Disabled Dance-Maker." In *Out of Time?*, edited by Backhausen E., W. Benjamin, and W. Noa, 193–202. Abingdon: Routledge.
Merleau-Ponty, M. 1968. *The Visible and the Invisible: Followed by Working Notes*. Evanston: Northwestern University Press.
Roesler, E., M. Heuring, and L. Onnasch. 2023. "(Hu)man-Like Robots: The Impact of Anthropomorphism and Language on Perceived Robot Gender." *International Journal of Social Robotics* 15:1829–1840. https://doi.org/10.1007/s12369-023-00975-5.
Rogel, A., R. Savery, N. Yang, and G. Weinberg. 2022. "Robogroove: Creating fluid motion for dancing robotic arms." In *Proceedings of the 8th International Conference on Movement and Computing*, 1–9.
Savery, R., A. Rogel, and G. Weinberg. 2024. "Robotic Dancing, Emotional Gestures and Prosody: A Framework for Gestures of Three Robotic Platforms." In *Sound and Robotics*, edited by R. Savery, 285–316. New York: Chapman and Hall/CRC.
Sennett, R. 2008. *The Craftsman*. London: Yale University Press.
Shiomi, M., K. Shatani, T. Minato, and H. Ishiguro. 2018. "How Should a Robot React Before People's Touch?: Modeling a pre-Touch Reaction Distance for a Robot's Face." *IEEE Robotics and Automation Letters* 3 (4): 3773–3780. https://doi.org/10.1109/LRA.2018.2856303.
Wenger, E. 1998. "Communities of Practice: Learning as a Social System." *Systems Thinker* 9 (5): 2–3.
Whatley, S. 2023. "Dance, Disability, and Keeping "In Time" on Strictly." In *Out of Time?*, edited by E. Backhausen, B. Wihstutz, and N. Winter, 61–78. Abingdon: Routledge.
Whatley, S., S. Benford, K. Marsh, F. Zhou, I. Ashcroft, P. Tennent, W. O'Brien, and T. Erhart. 2023. "Personalising Prosthetics: Digital Interventions in Disability and Dance." *International Journal of Performance Arts and Digital Media* 19 (3): 318–333. https://doi.org/10.1080/14794713.2023.2196884.
Yang, L. 2022. "Influence of Human–Computer Interaction-Based Intelligent Dancing Robot and Psychological Construct on Choreography." *Frontiers in Neurorobotics* 16:819550. https://doi.org/10.3389/fnbot.2022.819550.
Zheng, X., M. Shiomi, T. Minato, and H. Ishiguro. 2020. "How Can Robots Make People Feel Intimacy Through Touch?" *Journal of Robotics and Mechatronics* 32 (1): 51–58. https://doi.org/10.20965/jrm.2020.p0051.
Zhou, F., S. D. Benford, S. Whatley, K. Marsh, I. Ashcroft, T. Erhart, W. O'Brien, and P. Tennent. 2023. "Beyond Skin Deep: Generative Co-design for Aesthetic Prosthetics." In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 1–19.